

Modeling Climate Anomalies
with Statistical Analysis

Today's goals



Module 1

Introduction to Python for Data Analysis

Course Introduction

- ✓ Course Updates and Accessibility Support Reading • 1 min
- ✓ Earn Academic Credit for your Work! Reading • 10 min
- ✓ Course Support Reading • 10 min
- Introduction to the Course Video • 2 min
- Meet the Instructor Video • 1 min
- ✓ Introduction to Pandas for Data Loading and Exploration Video • 25 min
- ✓ Pandas Reading • 15 min
- ✓ Course Assignments with Jupyter Reading • 10 min

Visualizing Data with Python

- ✓ Introduction to Matplotlib and Seaborn for Data Visualization Video • 25 min

Module Assignment

- ✓ Introduction to Pandas and Matplotlib Programming Assignment • Grade: 100%
- ✓ Pandas and Matplotlib Graded Assignment • Grade: 93.33%

Module 2

Collecting Climate Data

Data Collection and APIs

- ✓ Introduction to APIs Video • 9 min
- ✓ APIs, JSON, and Python Requests Reading • 15 min
- ✓ API Hands-On (Lab Activity) Reading • 30 min
- API Hands-On Graded Assignment • 15 min

Data Sources: NOAA

- Introduction to NOAA Website for Data Collection Video • 9 min
- NOAA API and Available Data Reading • 10 min

Data Sources: USGS

Introduction to USGS Website and Available Data

API Hands-On (Lab Activity)



0:32 / 1:37



2x



In this lab, you will learn how to use the Python requests library to make API calls. You will practice retrieving data from a public API, processing the response, and extracting useful information. This exercise will help you understand how to interact with web services programmatically and integrate external data into your Python projects.

After completing the lab, you will take a short quiz to assess your understanding of making API calls with Python and interpreting the results. The quiz will include questions about the API you used, the data you retrieved, and the techniques you implemented for error handling and data processing. Ensure you have your code and results ready, as you may need to reference them during the quiz.

For this lab you will need Jupyter and the Python library `requests` installed.

Instructions

[Data USA](#)  (external link; opens in a new tab) pulls together public US Government data into an open, easy-to-use platform.

1. For the first exercise make an api call to the following url and print the response status code. Once you have printed the request status code, [review your code and other codes](#)  (external link; opens in a new tab) to see what your code means.
2. Once you have a successful API call, use the `json()` method to fetch data from the API and print the data. Note the returned json contains two names, data and source; you should only print the values from data.
3. Finally, use your json data to create a pandas dataframe using the `json_normalize()` method.

To help get you started, here is a lab file with some starter code.



Assignment 2 _ API Practice.ipynb

ZIP File

Go to next item

✓ Completed



Like



Dislike



Report an issue

Go to next item →